Advanced International Macroeconomics B (EI136)

Class 10

Artificial intelligence

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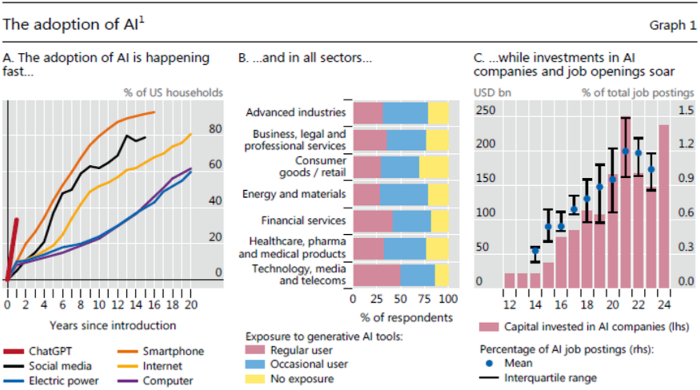
Outline of today class

- How does AI (higher productivity) impact output and inflation (Aldasoro, Doerr, Gambacorta and Rees 2024)?
  - Pattern of expectations, sectoral composition of AI effect.
- Contrasted impact between advanced and emerging economies (Gambacorta, Kharroubi, Mehrotra, and Oliviero 2026).
- Interaction between AI and demand, possibility of recession (Fornaro and Wolf 2026).
  - Shift of income away from workers reduces demand.
  - Short run inflation - employment trade-off, long run risk of zero lower bound.
- Challenges looking forward (Korinek 2024).
- Taxation of robots (Thuemmel 2023).
  - Different types of labor with different robot exposure.
  - Subsidize robots at first, then tax. Robot tax is very secondary compared to optimal labor taxation.

IMPACT OF AI

Rapid transformation

- AI is adopted faster than earlier technologies (easier to use), more broadly, with big investments.



Bank for International Settlements (2024). "Artificial intelligence and the economy: implications for central banks" BIS annual report.

AI exposure

- Model AI as a productivity improvement (+ 1.5% TFP growth), and assess sectoral linkages (Aldasoro, Doerr, Gambacorta and Rees 2024).
    - AI can be used rapidly: no need for new business practices, or new equipment.
    - AI raises productivity of workers with cognitive tasks, and raises innovation - hence future productivity.
- Compute the exposure of AI across industries.
    - Relevance of 10 AI applications to 52 workplace abilities (oral communication, deductive reasoning, stamina,...). Score of 0-10.
    - Combine 52 abilities with occupations, to get a AI exposure of occupation.
    - Use share of occupations in each industry to construct AI exposure of industry.
    - Highest AI exposure: finance and insurance, management.

AI in a macro model

- Multisector DSGE model. 20 sectors, with each used as input in other sectors through CES aggregator.
  - Rich connections across sectors, calibrated on input-output database.
- AI raises current and future productivity, moves the steady state.
- Linearize the model, but allows for time-varying coefficients linking the variables.
- Two scenarios of how agents anticipate future movements in coefficients:
  - Unanticipated: changes happening until each period are understood, but subsequent changes are not expected (uncertainty on the impact of AI).
  - Anticipated: subsequent changes are fully understood and factored in.

Small model 1: K vs. L

- 3 industries: a labor intensive one, a capital intensive one, and an intermediate one.
- Contrast impact of AI (+ 10% of productivity) in each industry alone, and impact of broad AI (+ 3.3% in productivity).
- All scenarios leads to a similar increase in GDP.
- Broad AI scenario raises the real wage, and the relative price of the labor intensive industry.
  - Output increases by more in capital intensive industry, as it is more price competitive.
  - Higher competitiveness raises capital return, and capital intensive industry is most able to benefit from extra capital.

Role of factor intensity

- Impact of AI depends on reaction of relative prices.

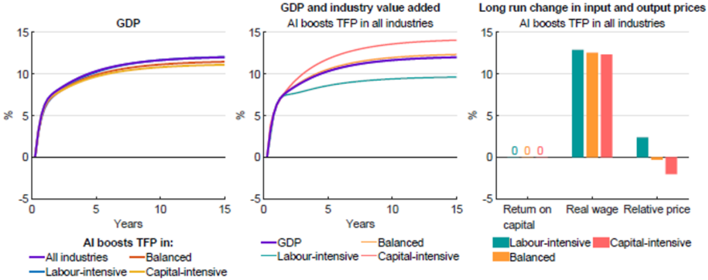


Figure 3: Simple model 1 – Cross-industry differences in production technologies

Aldasoro, Inaki, Sebastian Doerr, Leonardo Gambacorta, and Daniel Rees (2024). "The impact of artificial intelligence on output and inflation" BIS working paper 1179.

Small model 2: use of goods

- Three sectors with different use of their product: consumption good, capital good, intermediate good.
- GDP increase by most when AI is in consumption good, and by least if AI is in investment good.
- AI in consumption good sector:
  - Large increase in the other two sectoral prices.
  - Even increase in real CPI wage, real PPI wage increase much more in consumption sector.
  - Consumption sector reduces labor, which can go in the other sectors.
  - Raises production in the other sectors, which are inputs for the consumption sector.
- AI in investment good sector.
  - Large decrease of that sector's price.
  - Even (small) increase in real CPI wage, real PPI wage increase much more in investment sector.
  - Labor goes away from the investment sector, but the others (esp. consumption) are not inputs to the investment sector.

Role of place in value chain

- More impact when AI affects sectors at the end of the chain (consumption), freeing labor to go in sector at the beginning of the chain.

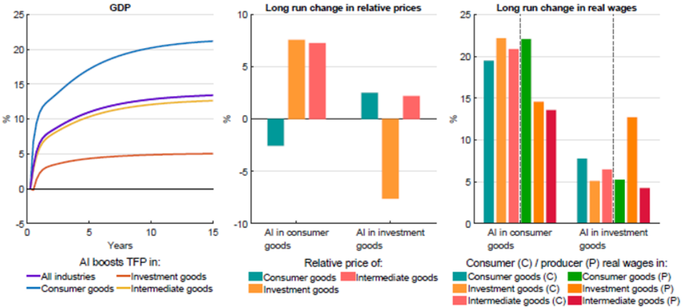
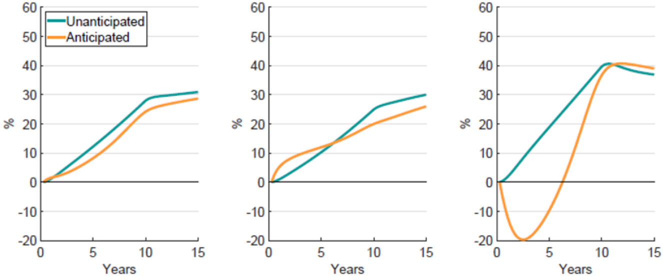


Figure 4: Simple model 2 – Differences in the use of industry output

Aldasoro, Inaki, Sebastian Doerr, Leonardo Gambacorta, and Daniel Rees (2024). "The impact of artificial intelligence on output and inflation" BIS working paper 1179.

Anticipated vs. unanticipated: GDP

- Broad model, contrast effect depending on expectations.
- Similar effect on GDP. Consumption is more front-loaded (at the expense of investment) in anticipated case, due to wealth effect.



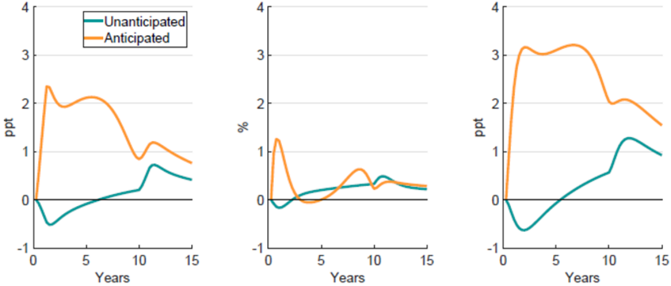
(a) GDP (b) Consumption (c) Investment

Figure 5: **Output, consumption, and investment**

Aldasoro, Inaki, Sebastian Doerr, Leonardo Gambacorta, and Daniel Rees (2024). "The impact of artificial intelligence on output and inflation" BIS working paper 1179.

Anticipated vs. unanticipated: inflation

- Unanticipated: supply effect dominates, lowering inflation. Anticipated: demand dominates, raising inflation.
- Policy responds quite different.



(a) Inflation (b) Output gap (c) Policy rate

Figure 6: **Inflation, output gap and policy rate**

Aldasoro, Inaki, Sebastian Doerr, Leonardo Gambacorta, and Daniel Rees (2024). "The impact of artificial intelligence on output and inflation" BIS working paper 1179.

Role of AI exposure and concentration

- AI leads to heterogeneous increase in industries' value added.
- Little correlation with AI exposure, strong negative correlation with labor share and with increase in relative price (similar to small model 1).
  - Relative prices matter more than AI exposure per se.
- Results similar if AI is labor (or capital) augmenting rather than general technology.
- If AI increase is concentrated in some sectors, magnitude of GDP increase is different.
  - Largest for AI in healthcare or manufacturing (inflation lowest if AI in manufacturing).
  - Lowest for AI in construction or professional services

ADVANCED VS. EMERGING ECONOMIES

Sectoral and policy heterogeneity

- Sectors differ according to their exposure to AI. Larger for finance, health care, education.
- Countries differ according to:
  - Sectoral composition.
  - Preparedness to AI: quality of digital infrastructure, regulation, human capital.
- What is the impact of preparedness at the sectoral and national level (Gambacorta, Kharroubi, Mehrotra, and Oliviero 2026)?
  - Measure of AI exposure of sectors (AIIE) and of AI preparedness of countries (AIPI).
- Effect on value added growth in 2022-23 (short sample).
  - One standard deviation increase of AIPI raises output, especially for high AIIE sectors.
  - It raises GDP more in advanced economies (around 0.6 percentage point).

- Heterogeneity of AI exposure of sectors, and sectoral composition of countries.

Figure 1. Industry-level exposure to AI

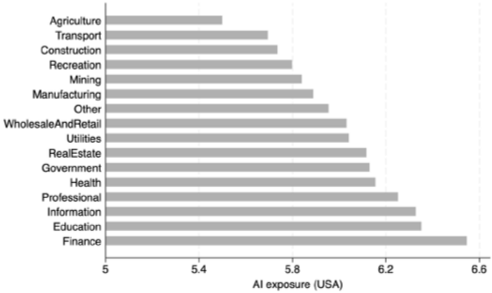
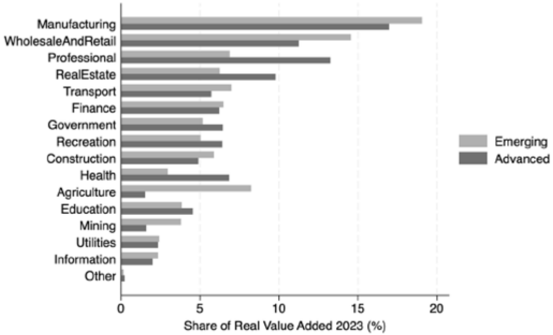


Figure 2. Share of industries in total real value added: AEs vs EMDEs

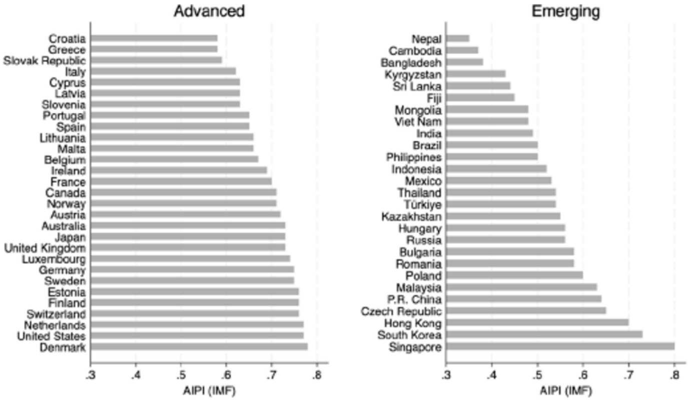


Gambacorta, Leonardo, Enisse Kharroubi, Aaron Mehrotra, and Tommaso Oliviero (2026). "Artificial Intelligence and Growth in Advanced and Emerging Economies: Short-Run Impact", CEPR discussion paper 20992

Preparedness

- Higher AI preparedness in advanced economies.

Figure 4. AIPI index (IMF): AEs vs EMDEs



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Econometric estimate

- Growth is higher in sectors with more AI exposure, especially in high preparedness countries.
- One standard deviation of AIPI raises growth by 16.5 ppt in sector at 10th percentile of exposure and 18.5 ppt at 90th percentile.

Table 1. Regression results, baseline model

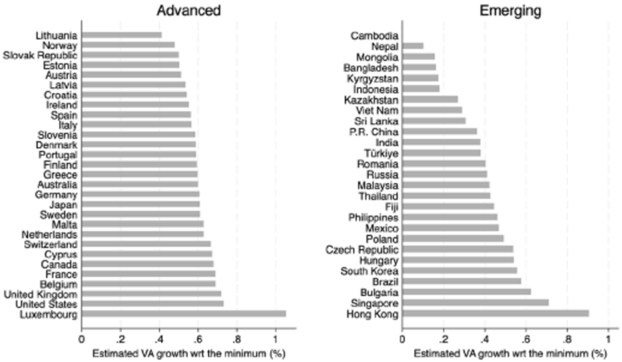
Dependent variable: Growth rate of real value added over 2022-2023			
	(I)	(II)	(III)
AI exposure	2.519*	-13.170*	
	(1.457)	(7.652)	
AI exposure X AIPI		2.905**	3.043**
		(1.352)	(1.394)
Share of 2022 Total Value Added	-0.134**	-0.160**	-0.186*
	(0.066)	(0.070)	(0.099)
Observations	875	875	875
R-squared	0.157	0.160	0.195
Industry FE	N	N	Y
Country FE	Y	Y	Y

Notes: OLS estimation with standard errors clustered at country level in parentheses. Significance level: *p<0.1; ** p<0.05; *** p<0.01.

Gambacorta, Leonardo, Enisse Kharroubi, Aaron Mehrotra, and Tommaso Oliviero (2026). "Artificial Intelligence and Growth in Advanced and Emerging Economies: Short-Run Impact", CEPR discussion paper 20992

- Combine sectoral estimate with sectoral composition.
- One standard deviation of AIPI raises aggregate growth more in advanced economies.

Figure 5. Estimated impact of AI on value added: AEs vs EMDEs



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AI AND DEMAND SLUMP

Impact in heterogenous economy

- Stylized model of the effect of AI in a context of heterogenous agents (Fornaro and Wolf 2026).
- Two types of agents.
  - Workers earn labor income, and do not save.
  - Capitalists earn capital income, and have wealth (capital and bonds) in the utility.
  - Lower propensity to consume of capitalists.
- Final good consists of a range of intermediate goods.
  - Goods $j > \overline{J}_t$ produced with labor.
  - Goods $j \leq \overline{J}_t$ produced with labor or capital, with linear technology. Assume that capital is used.
  - Production function of final good amounts to a Cobb-Douglas with labor share $1 - \overline{J}_t$.
- Downward wage rigidity can constrain demand.
- Tobin Q in capital accumulation.

AI shock

- Permanent increase of the share of goods $\overline{J}_t$ produced with capital.
  - Decrease in the labor share of final good production.
  - Shift of income from high propensity to consume workers to low propensity to consume capitalists.
- Impact with fixed capital stock. Broader role of capital raises its value.
  - Lower output share reduces consumption.
  - Higher value of capital raises wealth and consumption of capitalists.
  - First effect dominates and output falls.
- Impact with endogenous capital stock. Broader role stimulates investment.
  - Investment is also proportional to output (demand for capital). Low demand can reduce investment and potential future output.

- Permanent slump in the absence of reaction from the central bank (constant real interest rate, wages, never go down).
- Lower real rate stabilizes employment.

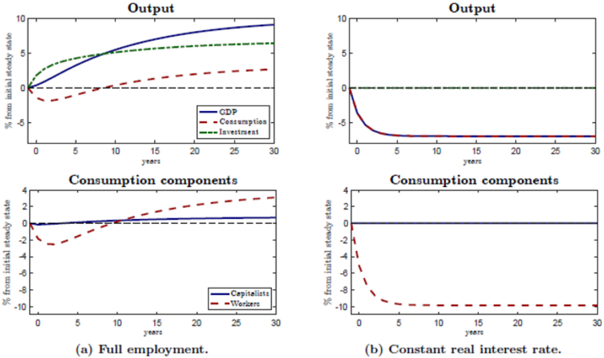

Figure 2: Impact of automation on output and consumption. Notes: Left column (a): Natural allocation. Right column (b): Constant real interest rate policy. The upper row decomposes output into its spending components: (aggregate) consumption and investment. Both are expressed in percentage deviation from the initial steady state, weighted by the initial steady state share in GDP. The lower row shows the components of aggregate consumption, namely worker consumption and capitalist consumption.

Fornaro, Luca, and Martin Wolf (2026). "Macroeconomic Policies for AI", CEPR discussion paper 21412.

Inflation trade-off

- Full employment requires a lower real wage (labor is less in demand), through a temporary inflation increase.

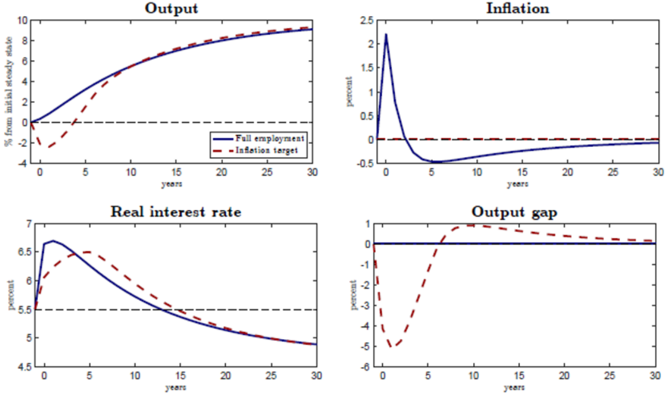

Figure 3: Monetary policy responses to advances in AI. Notes: Blue solid lines: Targeting full employment (natural allocation). Red dashed lines: Targeting constant inflation.

Fornaro, Luca, and Martin Wolf (2026). "Macroeconomic Policies for AI", CEPR discussion paper 21412.

Impact on natural interest rate

- r^* initially increases (higher investment) and then decreases (lower demand).
- Zero lower bound can bind ultimately. Can be offset by employment subsidy (with a slightly lower initial output increase).

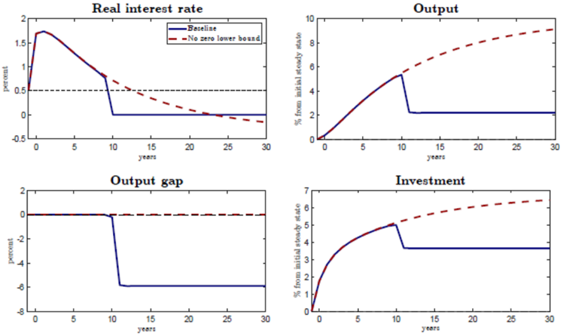

Figure 6: AI boom and liquidity trap. Notes: Blue solid lines: Targeting full employment as long as $r_t > 0$, set $r_t = 0$ else. Red-dashed lines: Targeting full employment, ignoring the zero lower bound.

Fornaro, Luca, and Martin Wolf (2026). "Macroeconomic Policies for AI", CEPR discussion paper 21412.

AI IMPLICATIONS

A profound change

- AI could be as big as transition from Malthusian to Industrial economy (Korinek 2024).
- Labor is not the bottleneck factor anymore.
  - Machines (AI) much more substitutable with labor than capital is.
  - Sharp deterioration of terms-of-trade for labor.
  - Artificial General Intelligence closer to human abilities than current AI, could be 5-10 years away.
  - Labor advantage concentrated in activities requiring trust and connection.
- Key skills (and teaching) will be ability to efficiently use AI.

Challenges for policy

- Eight major challenges.

1. Handle income concentration and inequality. Includes inequality across countries, as poorer ones may be less well equipped to use AI.
2. Reorient education towards AI literacy.
3. Beware of polarization between people using AI well and the others.
4. Macroeconomic implications: new forms of income (universal basic income), new drivers of inflation (not the labor slack any longer), new forms of taxation.
5. Antitrust with risk of extreme concentration (increasing returns to scale from data).
6. Intellectual property.
7. AI use of energy and environmental impact.
8. AI regulation, with global coordination.

TAXATION OF ROBOTS

Different labor - capital complementarity

- Adoption of robots has a different impact depending on the specific tasks of labor (Thuemmel 2023).
- Production using capital (robots), and three types of labor : cognitive, C, routine, R, manual, M.
  - Wage ranking: $w_C > w_R > w_M$.
  - Robots raises the marginal product of C and M more than the product of R.
- More robots raise wage inequality at the top (w_C moves above w_R), and reduces it at the bottom (w_M catches up to w_R).
- Consider the allocation by a planner with high weight on lower income.
  - Robots are initially good (w_M converges to w_R), but then bad once w_M exceeds w_R.
  - Subsidize robots at first, then tax them.

Calibrated model

- Consider different skills within each type of labor.
- Consider usual capital (equipment) with equal substitutability to all labor type, in addition to robots.
- Loss from robots concentrated among routine workers, most present at lower end of income scale.
- Cheaper robots lead to:
  - Higher capital and output.
  - Higher income of cognitive and manual workers.
  - Lower income of routine workers.
- Taxing robots reduces the inequality, but not by much.

- Impact of cheaper robots, with and without taxation.

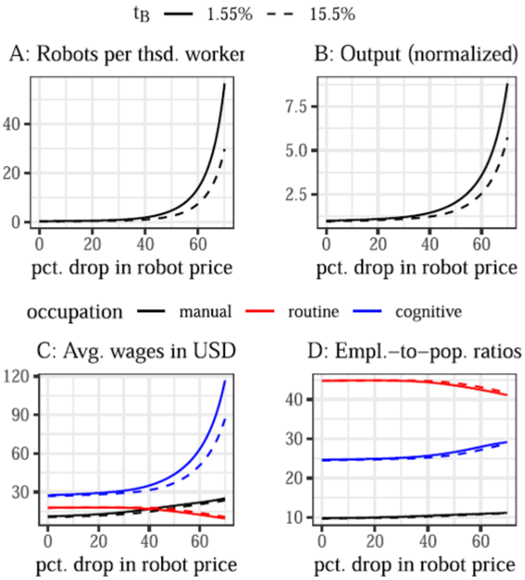

FIGURE 2. Effect of robot-price drop under status-quo and higher robot tax. The solid lines correspond to the status-quo tax system with a robot tax of 1.55%. The dashed lines correspond to a modified status-quo tax system with the only difference being a robot tax of 15.5%. Employment-to-population ratios represent the share of the population that participates in the labor market, categorized by occupation (not to be confused with the participation rate by occupation; summing the employment-to-population ratios gives the total participation rate).

Thuemmel, Uwe (2023) "Optimal Taxation of Robots", *Journal of the European Economic Association* 21(3), pp. 1154–1190

Optimal taxation

- Subsidize robots at first, then tax them. Tax on equipment capital is steady.

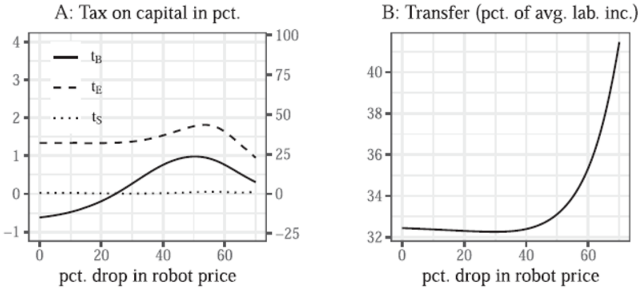

FIGURE 3. Optimal capital taxes and transfer. In the left panel, t_B: tax on robots, t_E: tax on equipment, and t_S: tax on structures; left axis: tax on capital stocks (as in the model) and right axis: tax on capital income flows, assuming an annual return of 4.36% (see Section 5.2 for the reverse conversion from taxes on flows to taxes on stocks).

Thuemmel, Uwe (2023) "Optimal Taxation of Robots", *Journal of the European Economic Association* 21(3), pp. 1154–1190

Implementable taxes

- Consider 4 reforms of taxes, starting from the current situation.
  - Reform 1: Optimal non-linear income tax, unchanged capital tax.
  - Reform 2: reform 1 + optimal capital tax on both robots and equipment capital.
  - Reform 3: reform 2 + different taxes on robots and equipment capital.
  - Reform 4: optimal robot taxes, all other taxes unchanged.
- Reform 1 improves welfare, but beyond this fine-tuning the taxation of capital and robots matters little.

Gains from reforms

- Optimal income tax is what matters.

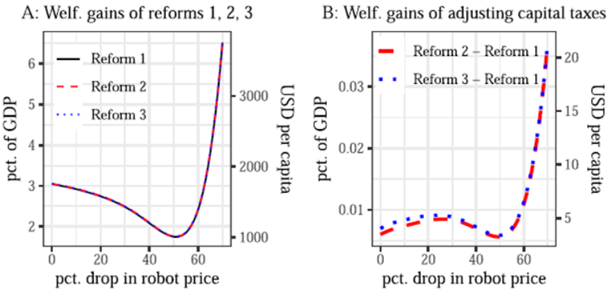

FIGURE 5. Welfare effects of tax reforms. In Panel A, the lines represent the welfare-equivalent monetary value of the three tax reforms. Panel B shows the difference in these values between selected reforms, thereby capturing the gains from adjusting capital taxes. For example, Reform 2 – Reform 1 captures the value of setting capital taxes optimally under the restriction that equipment and robots must be taxed at the same rate.

Thuemmel, Uwe (2023) "Optimal Taxation of Robots", *Journal of the European Economic Association* 21(3), pp. 1154–1190

Effect of tax on robots

- Minimal when the other taxes are left unchanged.

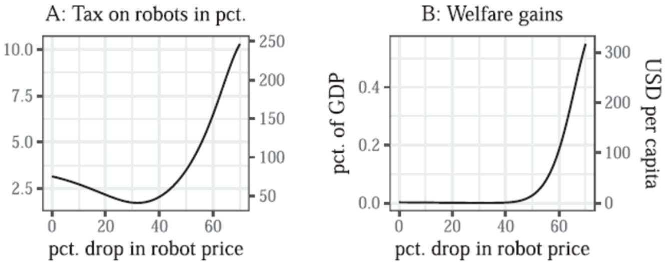

FIGURE 6. Optimal robot tax and welfare gains for Reform 4. Panel A, left axis: tax on capital stocks (as in the model); right axis: tax on capital income flows, assuming an annual return of 4.36% (see Section 5.2 for the reverse conversion from taxes on flows to taxes on stocks). In panel B, the line represents the welfare-equivalent monetary value of adjusting the robot tax.

Thuemmel, Uwe (2023) "Optimal Taxation of Robots", *Journal of the European Economic Association* 21(3), pp. 1154–1190